

FTC Autonomous Teacher Lesson Plan

Part 11 - Build an Accurate L Shaped Path

Middle and high school FTC students | 75 to 90 minutes | Teams of up to eight

Students combine the encoder-distance method from Part 9 with the IMU-turn method from Part 10. They command a 36-inch first leg, a 90-degree right turn, and a 24-inch second leg while using heading feedback to correct drift.

Lesson outcome

Students can assign distance and direction to the correct sensors, trace three readable autonomous commands, explain heading correction during straight travel, and evaluate final position using evidence from an L-shaped floor course.

Learning objectives

- Explain that encoders estimate distance while the IMU measures heading.
- Read a reusable driveStraight distance and target-heading method call.
- Trace drive 36 inches at 0 degrees, turn to -90 degrees, and drive 24 inches at -90 degrees.
- Calculate heading error and predict the left-right motor correction.
- Explain encoder targets, RUN_TO_POSITION, normalized heading error, timeout, and explicit stop.
- Measure distance, angle, sideways error, and endpoint error over repeated trials.
- Perform the POWERING ON procedure for a multi-segment floor path.

Required materials

- [] Calibrated two-motor goBILDA StarterBot used in Parts 9 and 10
- [] Rigidly mounted Control Hub, Driver Station, charged battery, and StarterBotLPathAutonomous.java
- [] At least an eight-foot-square clear floor area
- [] Painter's tape, 12-foot tape measure, framing square, and floor-safe labels
- [] Start, corner, finish, side-boundary, and stopping-buffer markers
- [] Printed trial record, lesson plan, and POWERING ON safety poster

Teacher preparation

- [] Confirm Part 9 distance calibration and Part 10 Hub-orientation settings are recorded.
- [] Confirm hardware names are left_drive, right_drive, and imu.
- [] Inspect drivetrain alignment, wheel retention, encoder cables, Hub mounting, and battery condition.
- [] Build the Java file and confirm the Autonomous OpMode appears on the Driver Station.
- [] Tape a 36-inch first leg, a square 90-degree corner, a 24-inch second leg, and extra stopping space.
- [] Mark one consistent robot reference point for every distance and endpoint measurement.
- [] Select low starting power and verify every stop path before class.

Mission checkpoints

CHECKPOINT	DISTANCE COMMAND	TARGET HEADING	EXPECTED STATE
Start	0 inches	0 degrees	Robot square, yaw reset, motors stopped
End of leg 1	36 inches	0 degrees	At corner line, facing original direction
After turn	Turn only	-90 degrees	Inside heading tolerance, motors stopped
End of leg 2	24 inches	-90 degrees	At finish mark, facing right
Safe approach	None	Reported only	STOP pressed and DISABLED confirmed

Sensor responsibilities

QUESTION	PRIMARY EVIDENCE	WHY
How far should the wheel turn?	Encoder counts	Counts are calculated from wheel and drivetrain geometry.
Which way is the robot facing?	IMU yaw	Yaw reports heading around the vertical robot axis.
Is the robot drifting during a straight leg?	Heading error	Target heading minus current yaw reveals directional error.
Did the robot reach the floor mark?	Physical measurement	Sensors estimate motion; the taped course provides external evidence.

Student roles

ROLE	RESPONSIBILITY
Safety captain	Controls the complete L-shaped boundary and authorizes INIT and START.
Driver Station operator	Runs the OpMode and remains ready to press STOP.
Course marshal	Squares the disabled robot and removes all setup tools before POWERING ON.
Code reader	Traces the three method calls and identifies each sensor's job.
Distance measurer	Measures both legs from the agreed robot reference point after DISABLED.
Angle observer	Records turn angle and final heading after safe entry is authorized.
Data recorder	Records telemetry, floor measurements, trial conditions, and exit reason.
Debugging analyst	Proposes one evidence-supported mechanical or software change.

Nonnegotiable safety boundary

The complete first leg, turn sweep, second leg, side boundaries, and stopping buffer must remain empty while the robot is enabled. Measuring equipment must be removed before POWERING ON. Students approach only after STOP is pressed, DISABLED is visually confirmed, and the safety captain opens the boundary.

POWERING ON for the L path

1. Place the robot squarely at the start while the Driver Station shows DISABLED.
2. Remove hands, tools, cables, tape measures, and setup equipment from both travel legs.
3. Check the first leg, complete turn sweep, second leg, side boundaries, and stopping buffer.
4. Call loudly: POWERING ON!
5. Pause while everyone moves behind the boundary.
6. Visually confirm the complete L-shaped route is clear.
7. Safety captain authorizes INIT and START; the operator remains ready to press STOP.
8. After motion, press STOP and confirm DISABLED before anyone measures or touches the robot.

Abort immediately if

- The robot moves in an unexpected direction or leaves the marked boundary.
- A person, tool, cable, or measuring device enters the course.
- A wheel, battery, wire, Control Hub, or mechanism appears loose.
- Heading or encoder telemetry is missing, frozen, or clearly unreasonable.
- The robot oscillates, stalls, or continues after the expected timeout.

Safety captain call

Use consistent words: 'Route clear?' Wait for visual confirmation. Call 'POWERING ON!' Wait again. Then say 'INIT approved' and 'START approved.' After the operator presses STOP and verifies the screen, call 'DISABLED confirmed; course open.'

Video pause guide

Use these planned pause points when the Part 11 video is available. Minor timestamp adjustments may be needed after final editing.

PAUSE	TEACHER QUESTION	EXPECTED IDEA
0:10	What complete path will the robot attempt?	Forward 36 inches, right 90 degrees, forward 24 inches, stop.
0:21	Which sensor is responsible for distance?	Motor encoders.
0:32	Which sensor is responsible for direction?	The Control Hub IMU and its yaw value.
0:43	Why do we pass both distance and heading to driveStraight?	One controls travel; the other controls directional correction.
0:54	What heading should leg 1 maintain?	Zero degrees.
1:05	Why use -90 rather than 'turn right some amount'?	It is an absolute measured target that telemetry can verify.
1:16	What heading should leg 2 maintain?	Negative 90 degrees.
1:28	Can equal commanded power guarantee a straight path?	No; mechanics, traction, and load can create drift.
1:40	If yaw is left of target, how should motor power change?	The correction steers the chassis back toward the target.
1:49	Which telemetry values help explain the result?	Targets, encoder positions, yaw, heading error, correction, and timeout.
1:57	When may the team enter the course?	Only after STOP, visible DISABLED, and safety-captain approval.

Lesson flow

STEP	ACTIVITY	TIME	TEACHER ACTION	STUDENT ACTION
1	Predict the L path	5 min	Show the taped path and ask which sensor solves each segment.	Label distance and heading checkpoints.
2	Watch with pauses	12 min	Use the pause guide while the robot remains disabled.	Annotate the three method calls and feedback loop.
3	Readiness audit	10 min	Verify Part 9 calibration, Part 10 orientation, and mechanical condition.	Inspect and initial each prerequisite.
4	Trace the sequence	10 min	Model driveStraight, turnToHeading, and the second driveStraight call.	Predict sensor targets and stop conditions.
5	Correction exercise	8 min	Give sample yaw errors and ask for left-right corrections.	Calculate error and predict steering direction.
6	First L-path trial	12 min	Enforce the complete floor boundary and POWERING ON sequence.	Perform roles and record all checkpoints.
7	Measure and diagnose	10 min	Separate distance, angle, endpoint, and repeatability errors.	Classify evidence before suggesting a change.
8	One change and retest	10 min	Approve one controlled adjustment.	Repeat the safety sequence and compare results.
9	Exit explanation	5 min	Ask how encoders and IMU cooperate.	Submit an individual sensor-role explanation.

Success criteria

- The robot completes the three commands in the intended order.
- Both drive legs remain close to their assigned headings.
- The turn finishes inside the agreed heading tolerance.
- The code stops after completion, timeout, Driver Station STOP, or unexpected exit.
- Students classify errors before changing code.
- No one enters the route before DISABLED is confirmed.

Hands on walkthrough

Course and readiness audit

The team tapes a 36-inch first leg, a square corner, and a 24-inch second leg. They add side boundaries and stopping space. Students confirm the distance calibration from Part 9, Hub orientation and yaw sign from Part 10, hardware names, encoder cables, wheel retention, and the common measurement reference point.

Teacher question: Which missing prerequisite would make the combined test uninterpretable?

Human code trace

Three students stand at the start, corner, and finish holding cards for `driveStraight 36 at 0`, `turnToHeading -90`, and `driveStraight 24 at -90`. A program token moves between them. At each station, the code reader identifies the target, sensor feedback, completion test, timeout, and motor stop.

Teacher question: Why should each helper stop the motors before the next helper begins?

Heading correction exercise

Use a paper robot and three yaw readings. For a zero-degree target, compare current headings of +6, -4, and 0 degrees. Students calculate target minus current, point in the needed steering direction, and indicate which motor should receive relatively more power. Repeat for a -90-degree target.

Teacher question: What happens if `DRIVE_GAIN` is too large?

First combined trial

The marshal squares the disabled robot and removes the square and tape measure. The safety captain checks the full route and leads `POWERING ON`. The operator runs the `OpMode` and calls `STOPPED` only after pressing `STOP` and visually confirming `DISABLED`. The team then measures leg 1, turn angle, leg 2, sideways offset, and endpoint error.

Teacher question: At which checkpoint did the largest error first appear?

Classify before changing

Classify the result as distance scale error, heading error, repeatability problem, mechanical problem, or course-setup problem. Choose one variable or mechanical correction, predict its effect, rerun the full safety sequence, and compare the second trial with the first.

Teacher question: Does the evidence support changing counts per inch, turn gain, drive gain, maximum power, or the robot mechanics?

Technical notes

Readable mission. Keep the top-level sequence short: driveStraight 36 at 0, turnToHeading -90, driveStraight 24 at -90. Put sensor details inside reusable helpers.

Encoder targets. Each drive leg adds calculated counts to the motors' current encoder positions and uses RUN_TO_POSITION. Power remains a positive magnitude for forward targets.

Heading hold. During each drive loop, target heading minus current yaw produces a steering correction. Clip correction so it cannot reverse a drive side unexpectedly at beginner speeds.

Motor mixing. With mirrored motor directions already configured, relatively more left power steers right and relatively more right power steers left while traveling forward.

Normalized angles. AngleUnit.normalizeDegrees prevents a large wrong-way correction when headings cross the -180 to +180 boundary.

Independent timeouts. Each drive leg and turn receives its own timer. A stalled first segment must not consume the safety limit intended for later segments.

Telemetry. Report current segment, encoder targets and positions, target heading, current yaw, heading error, correction, elapsed time, and exit reason.

Error diagnosis. Counts-per-inch changes address consistent distance scale error. Heading gains address directional response. Large inconsistent errors require mechanical and setup inspection first.

Stop behavior. Normal completion, timeout, Driver Station STOP, and finally must all command zero power and return the motors to a known mode.

Assessment evidence

- Correct sensor assignment for distance and heading
- Accurate trace of all three top-level commands
- Correct heading-error and correction predictions
- Unprompted completion of the full L-path safety sequence
- Complete checkpoint and endpoint measurements
- Reasonable error classification and one-variable retest

Exit prompt

Explain why encoder targets alone do not guarantee that the robot follows the taped L shape. Describe how IMU heading feedback changes the motor commands during a straight leg.